

$$\rho = \sum_{a=0}^{n-1} p_a |\phi_a\rangle\langle\phi_a| \quad p_a \neq 0$$

$$|\psi\rangle = \sum_{a=0}^{n-1} \sqrt{p_a} |\phi_a\rangle \otimes |a\rangle$$

Any set of orthonormal vectors

$$\text{Tr}_Y(|\psi\rangle\langle\psi|) = \rho$$

Ex. $\rightarrow \rho = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1|$

$$|\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle \otimes |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \otimes |1\rangle$$

→ Cryptography

→ It is impossible to implement an ideal form of bit commitment using Quantum Information

Refer GitHub
handwritten notes
utsavkaria-tech

